

Moreover, one can now use either Darktable or RawTherapee as GIMP plugins for opening raw files. Any recent version of either application will do. A new ‘Clip Warning’ display filter will visualise under-exposed and over-exposed areas of a photo, with customisable colours. For now, it is mostly geared towards images where colours are stored with floating point precision. You will benefit from this if you work on 16-/32-bit per channel float images such as EXR and TIFF.

Plugins

GIMP now ships with over 80 GEGL based filters. A lot of these are former GIMP effects. Here are some reasons why GEGL based implementations are better:

- One can apply them on images in 32-bit per colour channel precision mode.
- One can preview them right on the canvas, and if an image is larger than the viewport, GIMP will render the viewport first for immediate feedback.
- One can use split preview to compare an original image with its processed version, and swap before/after sides both horizontally and vertically.
- In a future, non-destructive GIMP, one will be able to adjust settings of those filters without undoing a ton of steps.

Some of the GEGL based filters have the OpenCL version for hardware acceleration. This will come in handy if OpenCL drivers work well for you. Besides, many operations can be multi-threaded, using the power of your processors to their full extent.

While working with active users, quite a few usability issues were got rid of. Here are just some of these changes:

- All transformation tools now automatically disable original layer view so that you can clearly see adjustments against the backdrop.

- Masks can now easily be created with the last values you used by just pressing *Shift* and clicking on the respective layer’s preview.
- All dialogue boxes except the ones like Scale now remember the last values you used across sessions.
- All GEGL based filters allow saving named presets and automatically make timestamped presets for the last time you used them.
- You can now choose the fill colour or pattern for empty spaces after resizing the canvas.

File format support

GIMP is now capable of reading and writing TIFF, PNG, PSD and FITS files with up to 32-bit per channel precision, where applicable. The PSD plugin additionally supports pass-through, hard mix, pin light, vivid light and linear light blending modes. GIMP now also ships with native WebP support, including features like animation, ICC profiles, and metadata. The JPEG 2000 plugin has been rewritten to use the OpenJPEG library rather than the somewhat obsolete Jasper library. Finally, the PDF plugin now supports importing password-protected files (one needs to know the password) and exporting multi-page PDF documents (each layer will be a page).

Metadata viewing, editing and preservation

GIMP now ships with plugins for viewing and editing Exif, XMP, IPTC, GPS and DICOM metadata. These are available via the *Image > Metadata* sub-menu. GIMP will also preserve existing metadata in TIFF, PNG, JPEG and WebP files. Each plugin has its options when exporting, to enable or disable exporting the metadata. Additionally, users can now set defaults for whether or not to preserve metadata in all affected file forms at plugins, at once. This depends on

whether they want complete privacy; if not, they can do a lot of micro-stock photography. The settings are available on the *Image Import and Export* page in *Preferences*.

On-canvas interaction

GIMP 2.10 ships with a new feature that allows some GEGL based filters to render on-canvas controls. For now, this applies to just three filters: Spiral, Supernova and Panorama Projection.

Search system

As GIMP keeps evolving and getting more features, it becomes increasingly difficult to locate particular commands in the user interface. This is especially true when one is a new user who has yet to master image editing or an experienced user who is accustomed to a different application. GIMP 2.10 comes with a search system that allows you to easily locate a command available in GIMP’s menu. The slash key (/) has to be pressed, a keyword typed and the command that looks most applicable is chosen.

The next big update will be v3.0, which will feature the GTK+3 port and a lot of internal changes. For users, this will mostly mean an updated user interface, better support for graphic tablets, better support for HiDPI displays, and better support for Wayland on Linux.

An overview of the image formats

Let’s take a look at the pros and cons of various image formats from a GIMP perspective.

XCF

The eXperimental Computing Facility (XCF) is where GIMP was authored.

Pros

- This is the native GIMP image format.
- Everything is saved—layers, selections, channels, paths and more.